Geomorphology I

Earth's Interior, Rocks and Plate Tectonics

Syllabus covered — GS-I: "Salient features of world's physical geography ... important geophysical phenomena such as earthquakes, tsunami, volcanic activity". Prelims: "Physical geography of India and the World".

What this topic is really about. The Earth's surface is not fixed. Its outer shell is broken into rigid plates that ride on a weak layer of hot mantle, and almost every large landform — fold mountains, ocean trenches, island arcs, rift valleys — and almost every big earthquake and volcano sits where two plates meet. This unit builds that one idea from the evidence up: how we know what is inside the Earth, what rocks are made of, and how the theory grew from continental drift to plate tectonics.

Study window	Mon 16 Nov 2026 — **Block A, 2:30–4:30** (1 slot)
Estimated study time	2 hr first pass, including the 10-minute D0 recall at the end
Revision time	15 min full note • 5 min one-page sheet (Section 16)
Study as	One chain of evidence. Learn Fig. 1 (interior) and Fig. 4 (plate boundaries) cold; most of the packet hangs off them.
Note status	**Draft 1 — for your review.** Not final until you sign it off.
Boundary	Earthquake measurement, volcano types, tsunamis and landforms are **G2** (17 Nov). India's physiography is **G7**. Earth–Sun system and the Juno question are **Map session M1** (21 Nov).

Candid verdict — thin in Prelims, steady in Mains. Prelims asks about this unit roughly once a year, and always on a narrow fact: P- vs S-waves (2023), the evidence for continental drift (2025), whether a rock is permeable (2025), which mountains are fold mountains (2024). Mains is different. GS-I has set **7 questions here in 14 years** (2013, 2014 ×2, 2018, 2020, 2022, 2025), almost all 10-markers, and the 2025 question was a 15-marker. Each one is answered with the same few diagrams. So the work here is concept plus diagrams for Mains, with a short list of Prelims traps on top. That is why it is Tier B and gets one slot, not two.

2 Scope map

What is included

- How we know the interior: direct and indirect sources, with seismic waves as the main evidence.
- The layered structure — crust, mantle, core — and the named discontinuities between them.
- Minerals and the three rock types, including the 2022 question on primary (igneous) rocks and the rock cycle.
- The chain of theory: continental drift → sea-floor spreading → plate tectonics, with the evidence for each.
- Plates and plate boundaries; the forces that move plates.
- Hotspots and mantle plumes — the 2018 question.
- Why fold mountains, earthquakes and volcanoes cluster at plate margins (2014); the Circum-Pacific belt (2020); island arcs of Indonesia and the Philippines (2014).
- How continents and ocean basins change shape over time — the Wilson cycle, for the 2025 question.
- India's plate journey: Gondwana, the Deccan and Rajmahal Traps, the Himalaya, the Andaman subduction zone and Barren Island.

Excluded material	Belongs to	Why it is kept out
Earthquake magnitude and intensity, seismic zones of India, volcano types, tsunamis	G2, 17 Nov	Named here only as plate-boundary effects
Weathering, erosion and landforms	G2	Exogenic processes; this unit is endogenic
Himalayan divisions, peninsular plateau, Deccan Trap resources	G7, 25 Nov	Here India appears only as a plate
Earth–Sun system, latitude and time; Juno mission (GS-I 2017)	Map session M1, 21 Nov	Astronomy, not geomorphology
Mineral resources and their distribution	G11, 10 Dec	Economic geography
Earthquake and tsunami disaster management	DM unit, 10 Dec	GS-III framework

How it connects outward

- **G2** — the hazards (earthquakes, volcanoes, tsunamis) are what plate boundaries produce. Read G1 first.
- **G6 Oceanography** — ocean-floor relief (ridges, trenches, abyssal plains) is the sea-floor-spreading story seen from above.
- **G7** — the Himalaya, the Indo-Gangetic trough and the Deccan Traps are all plate-tectonic products.
- **Anthropology 1.8** — geological time and dating methods you already know reappear in the age of ocean floors.
- **DM** — India's seismic vulnerability starts from the Indian plate's collision with Eurasia.

3 PYQ demand audit

Source: the UnlockIAS PYQ files you uploaded — Mains GS-I 2013–2026 verbatim, Prelims 1995–2026. The sub-topic tags are UnlockIAS's; the G1/G2 split is mine.

3A · Mains (GS-I)

Year	Question	Marks	Demand
2025	Discuss how the changes in shape and sizes of continents and ocean basins of the planet take place due to tectonic movements of the crustal masses.	15	Process — Wilson cycle
2022	Describe the characteristics and types of primary rocks.	10	Classification
2020	Discuss the geophysical characteristics of Circum-Pacific Zone.	10	Regional synthesis
2018	Define mantle plume and explain its role in plate tectonics.	10	Concept + role
2014	Why are the world's fold mountain systems located along the margins of continents? Bring out the association between the global distribution of fold mountains and earthquakes and volcanoes.	10	Cause + pattern
2014	Explain the formation of thousands of islands in the Indonesian and Philippines archipelagos.	10	Regional process
2013	(a) What do you understand by the theory of 'continental drift'? Discuss the prominent evidences in its support. (100 words)	5	Theory + evidence

What Mains is testing. Three kinds of question, each answered with a diagram. **Process** questions (2025, 2018) want a mechanism drawn step by step. **Pattern** questions (2014 ×2, 2020) want you to explain where something occurs — the answer is always a plate boundary. **Classification** questions (2022, 2013) want types with features and Indian examples. A Mains answer here without a diagram is a weaker answer.

3B · Prelims

Year	What was asked	The fact that decided it
2025	Evidence for continental drift: Brazil–West Africa rock belts, Ghana gold, Gondwana sediments	All three are textbook evidence (answer: all three)
2025	Chalk permeable, clay impermeable — why?	**Clay is porous but impermeable**; pores are too small to let water through
2024	Region–range–type matching (Vosges, Alps, Appalachians, Andes)	Vosges is a **block** mountain in Europe; Alps are fold
2023	P-waves arrive first; P is longitudinal, S is transverse	Both statements correct
2018	Barren Island volcano	Active, in Indian territory; it has erupted **after** 1991
2017	Closest to Great Nicobar	Sumatra — the same subduction arc
2013	Forces bringing dynamic change to the surface	All six: radiation, geothermal energy, gravity, plate movement, rotation, revolution
Pre-2011	Core is mainly iron (2009); magma is liquid + solid + gas, basaltic hotter than silicic (2001); sedimentary rocks form at the surface, contain fossils, occur in layers (2001); quartzite comes from sandstone (2001); India moved northward (1995); India was part of Gondwana (1995)	Use these for Sunday tests, not Friday sets

Trend reading. Prelims returns to this unit every year since 2023, but only for one question, and always a direct fact. Mains asks every two to four years and has moved from description (2013, 2014) to mechanism (2018, 2025). The next likely Mains question is another process question — how a hotspot track, a subduction zone or a rift evolves — so prepare mechanisms you can draw.

4 Big-picture conceptual map

MOVE 1 — We cannot see the interior, so we infer it. The deepest borehole reaches about 12 km of a 6,371 km radius. Almost everything else comes from how earthquake waves travel.

MOVE 2 — The interior is layered by density. Light crust floats on a dense mantle, around a very dense iron–nickel core. What matters for the surface is the weak **asthenosphere** under a rigid **lithosphere**.

MOVE 3 — The lithosphere is broken into plates that move. Wegener saw the continents had moved but could not say how. Sea-floor spreading supplied the mechanism; plate tectonics joined the two.

MOVE 4 — Boundaries do the work. Plates diverge, converge or slide past. Each type makes its own landforms and its own earthquakes and volcanoes.

MOVE 5 — Oceans open and close in cycles. The Wilson cycle explains how continents and ocean basins change shape and size — and why India now sits against Asia.

That chain is the shape of any long answer in this unit: **evidence → layers → moving plates → boundaries → cycles**.

5 Core notes

5.1 How we know what is inside the Earth

Plain version. Nobody has been there. We drill a little, we see what volcanoes bring up, and we listen to earthquake waves as they pass through.

Direct sources	Indirect sources
• **Mining and drilling.** The deepest hole, the Kola borehole in the Arctic, reached about 12 km. Ocean drilling projects sample the sea floor. • **Volcanic eruptions.** Magma and lava bring up material from depth.	• **Temperature, pressure and density** all rise with depth. • **Meteorites** — similar in make-up to the Earth's interior. • **Gravity anomalies** and the **magnetic field** (a liquid iron outer core generates it). • **Seismic waves** — the most important source. Their speed and path change with the material they pass through.

5.2 The layered structure

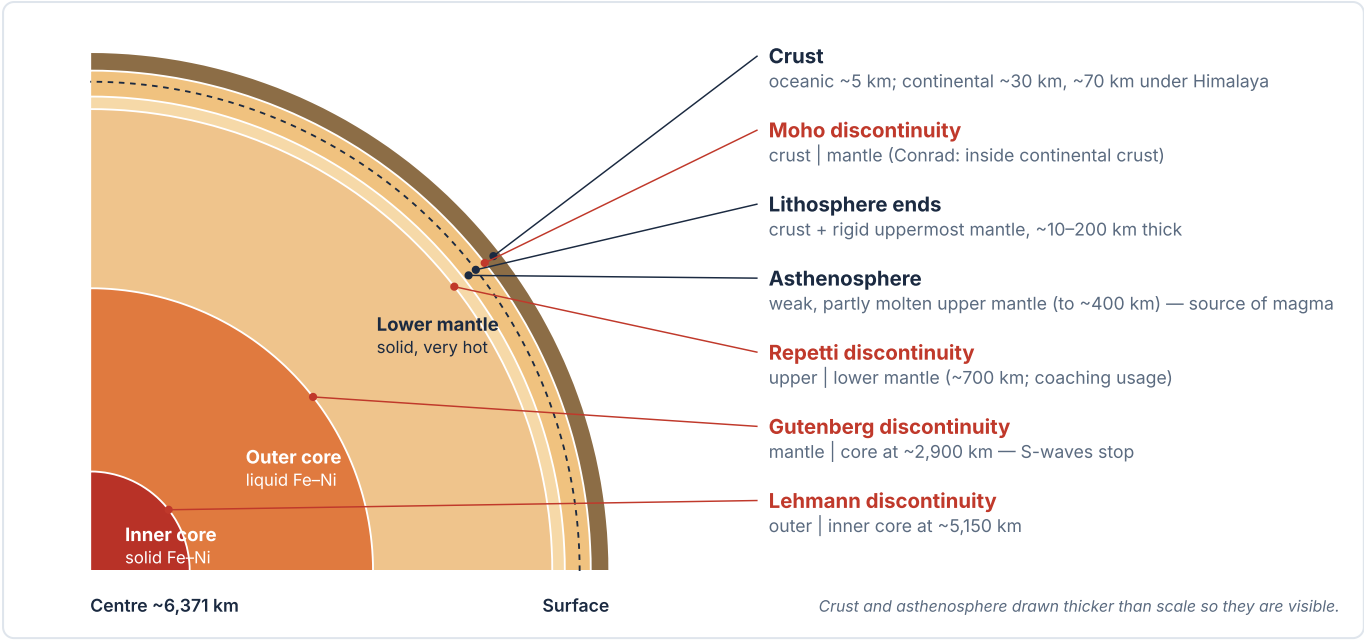


Fig. 1 — the Earth's interior with its discontinuities. This is the diagram for any interior question; draw it as a quarter-circle with labels on the right.

Layer	Depth	State and make-up	Key point
Crust	Oceanic ~5 km; continental ~30 km, ~70 km under the Himalaya	Solid, brittle. Continental crust is granitic (*sial*: silica + alumina); oceanic crust is basaltic (*sima*: silica + magnesium)	Thinnest layer; continental crust is thicker, lighter and much older
Mantle	Moho to ~2,900 km	Solid but able to flow slowly. The upper part includes the **asthenosphere** (to ~400 km), weak and partly molten	The asthenosphere is the source of magma and the layer plates slide over
Outer core	~2,900–5,150 km	Liquid iron–nickel	Movement here generates the magnetic field
Inner core	~5,150–6,371 km	Solid iron–nickel (pressure keeps it solid despite high temperature)	Densest part, around 13 g/cm³ at the centre

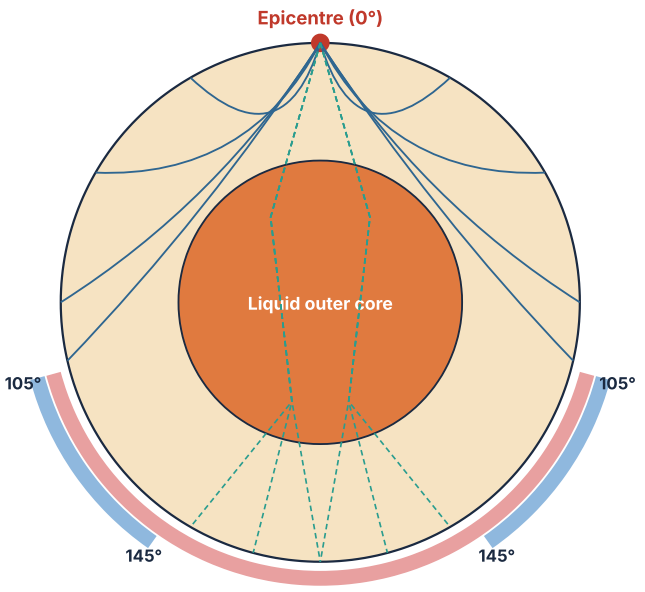
Lithosphere vs asthenosphere — the distinction that matters for plates. The **lithosphere** is the crust plus the rigid top of the mantle, about 10–200 km thick; it breaks into plates. The **asthenosphere** beneath it is weak and slowly flowing. Plates move because a rigid layer rides on a weak one. Do not confuse these with crust and mantle, which are divided by composition, not strength.

Discontinuities (boundaries where seismic speeds change sharply): **Conrad** — inside the continental crust, between granitic and basaltic layers; **Mohorovičić (Moho)** — crust | mantle; **Repetti** — upper | lower mantle, about 700 km (a coaching-standard name, rarely used in modern geology); **Gutenberg** — mantle | core at ~2,900 km; **Lehmann** — outer | inner core at ~5,150 km.

5.3 Seismic waves and the shadow zone

Wave	How particles move	Travels through	Speed and effect
P (primary)	To and fro along the direction of travel — longitudinal, like sound	Solids, liquids and gases	Fastest; arrive first
S (secondary)	At right angles to the direction of travel — transverse	**Solids only**	Slower; arrive second

Wave	How particles move	Travels through	Speed and effect
Surface waves (Love, Rayleigh)	Along the surface, side to side or rolling	Surface layers	Slowest; arrive last; **most destructive**



0°–105°: both P and S waves recorded

105°–145°: P-wave shadow zone (no direct P or S)

Beyond 105°: S-wave shadow — S never arrives

Beyond 145°: P arrives, bent (refracted) through the core

What the pattern proves
S-waves cannot travel through liquid. Their disappearance beyond 105° shows that the **outer core is liquid.**
P-waves slow and bend at the core boundary, creating the 105°–145° gap — the Gutenberg **discontinuity at ~2,900 km.**
Seismology is the main indirect source of everything we know about the interior.

Fig. 2 — the shadow zones. The S-wave gap beyond 105° is the evidence that the outer core is liquid.

5.4 Minerals and rocks

Composition of the crust. By weight: oxygen ~46.6%, silicon ~27.7%, aluminium ~8.1%, iron ~5.0%, calcium ~3.6%, sodium ~2.8%, potassium ~2.6%, magnesium ~2.1% (NCERT). Feldspar is the most common mineral group, then quartz.

A rock is an aggregate of one or more minerals. Rocks are grouped by how they formed.

	Igneous (primary)	Sedimentary	Metamorphic
How formed	Cooling and solidifying of magma or lava	Deposition and hardening of fragments, organic matter or precipitates	Existing rock changed by heat, pressure or fluids, without melting
Types	**Intrusive** (plutonic): cools slowly at depth, coarse crystals — granite, gabbro. **Extrusive** (volcanic): cools fast at the surface, fine or glassy — basalt, pumice, obsidian. **Acid** (silica-rich, light: granite) vs **basic** (silica-poor, dark: basalt)	**Mechanical** (clastic): sandstone, shale, conglomerate, loess. **Organic**: limestone, chalk, coal. **Chemical**: rock salt, gypsum	**Contact** (heat from nearby magma) and **regional** (pressure over large areas). **Foliated**: slate, schist, gneiss. **Non-foliated**: marble, quartzite
Features	Crystalline, hard, no layers, no fossils; host metallic ores	Layered (strata), may contain fossils, often porous; host coal, oil and gas	Recrystallised, often banded; harder than the parent rock
Indian examples	Deccan Traps (basalt), Rajmahal Traps; granites of the Peninsular shield	Vindhyan sandstone (Red Fort, Fatehpur Sikri); Gondwana coal (Damodar valley)	Makrana marble (Rajasthan); Dharwar schists (Karnataka)

Parent → metamorphic pairs (Prelims favourite): limestone → **marble** · sandstone → **quartzite** · shale → **slate** → phyllite → schist · granite → **gneiss** · coal → **graphite**.

Intrusive igneous landforms (bodies of magma that cooled underground): **batholith** (huge, deep), **laccolith** (dome-shaped), **lopolith** (saucer-shaped), **phacolith** (wavy, along folds), **sill** (horizontal sheet), **dyke** (vertical

sheet). Weathering of the Deccan basalt produces the black cotton soil.

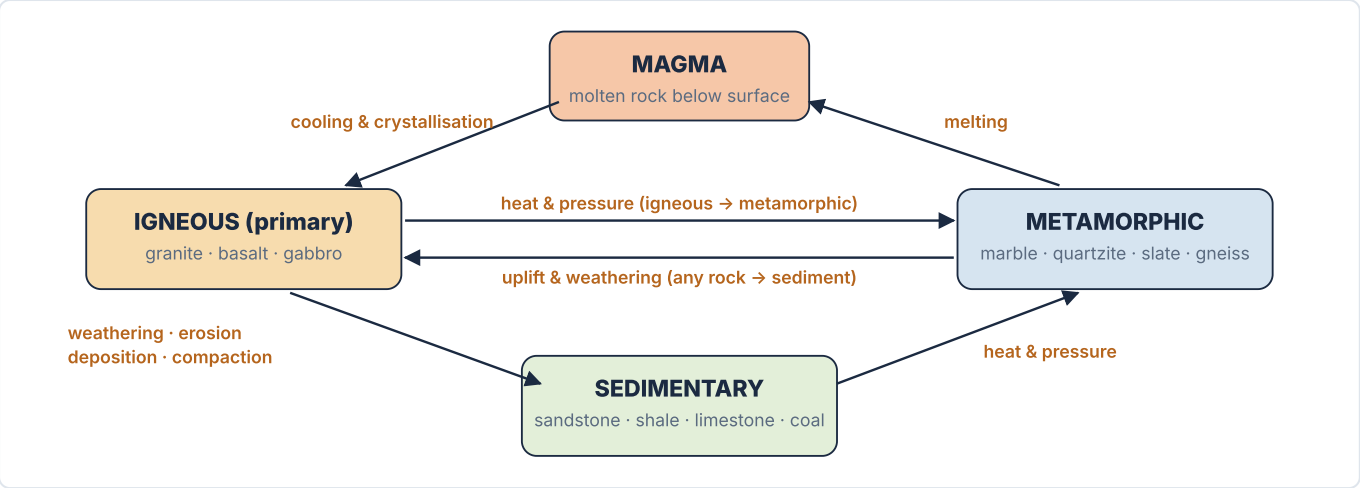


Fig. 3 — the rock cycle. Any rock can become any other; sedimentary rock covers most of the land surface but is a small share of the crust's volume.

Prelims trap — porosity vs permeability. *Porosity* is how much empty space a rock has; *permeability* is whether water can pass through. Clay is highly porous but nearly impermeable, because its pores are too tiny and poorly connected. Chalk and sandstone are permeable. The 2025 question turned on exactly this.

5.5 Continental drift — Alfred Wegener (1912–15)

Plain version. The continents fit together like a jigsaw, and the rocks and fossils on either side of the Atlantic match. Wegener said they were once one landmass that broke apart and drifted.

- **The idea.** One supercontinent, **Pangaea**, surrounded by one ocean, **Panthalassa**, began to split about 200 million years ago into **Laurasia** (north) and **Gondwanaland** (south), which broke further into today's continents.
- **Evidence.** (1) **Jigsaw fit** of South America and Africa — confirmed by Bullard's 1964 computer fit along the 1,000-fathom line. (2) **Rocks of the same age** across oceans — 2,000-million-year-old belts in Brazil match West Africa. (3) **Tillite** — glacial deposits; the Gondwana sediments of India have counterparts in six southern landmasses. (4) **Placer deposits** — Ghana's gold has no source rock in Ghana; the source is on the Brazil plateau. (5) **Fossils** — *Mesosaurus* in South Africa and Brazil only; *Glossopteris* flora across Gondwana; lemurs in India, Madagascar and Africa.
- **The forces he proposed.** A **pole-fleeing force** from the Earth's rotation, and a **tidal force** from the Sun and Moon. Both are far too weak.
- **Why it was rejected at first.** The evidence was strong; the mechanism was not. Arthur Holmes' idea of **convection currents** in the mantle (1930s), driven by radioactive heat, was the missing engine — but it was not accepted until sea-floor data arrived.

5.6 Sea-floor spreading — Harry Hess (1961–62)

Plain version. New ocean floor is made at the mid-ocean ridges and pushed outward, so the oceans widen and the continents move apart with them.

- **Post-war ocean mapping** found continuous **mid-ocean ridges** with volcanic activity along their crests, and deep **trenches** near some continents.
- **Evidence.** (1) Rocks are **youngest at the ridge** and older with distance on both sides. (2) The oldest ocean floor is **under 200 million years**, while continental rocks are up to about 3,200 million years old (NCERT) — oceans are recycled, continents are not. (3) Sediments are thin near ridges and thicker away from them. (4) **Magnetic stripes**: as lava cools it records the Earth's magnetic field, which reverses from time to time, so the sea floor shows matching stripes on both sides of the ridge (Vine and Matthews, 1963). (5) Earthquakes are shallow at ridges and deep at trenches.

- **Conclusion.** Crust is created at ridges and destroyed (subducted) at trenches, so the Earth does not grow in size.

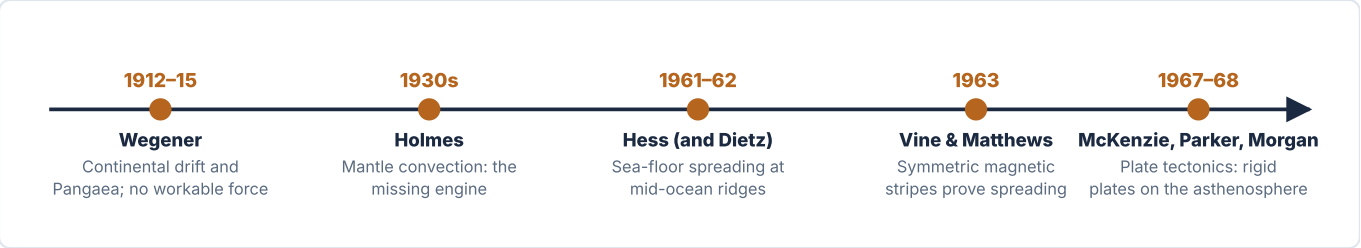


Fig. 7 — from drift to plate tectonics in five steps. Each step supplied what the previous one lacked.

5.7 Plate tectonics (1967–68)

Plain version. The lithosphere is cracked into about a dozen large rigid plates and many small ones. They move a few centimetres a year over the asthenosphere, and most geological action happens at their edges.

- **The theory.** Proposed independently by McKenzie and Parker (1967) and Morgan (1968). It joined continental drift and sea-floor spreading into one model.
- **Major plates** (NCERT list): Antarctic; North American; South American; Pacific; India–Australia–New Zealand; African; Eurasian. **Minor plates** include Cocos, Nazca, Arabian, Philippine Sea, Caribbean and Scotia.
- **Speeds.** From under 2.5 cm a year at the Arctic Ridge to over 15 cm a year at the East Pacific Rise (NCERT). The Indian plate moves north-east at about 4–5 cm a year today.
- **What moves the plates. Mantle convection** (Holmes' idea); **ridge push** (new crust at the elevated ridge slides away); and **slab pull** (the cold, dense sinking edge of a plate drags the rest behind it). Slab pull is generally thought to be the largest of the three.

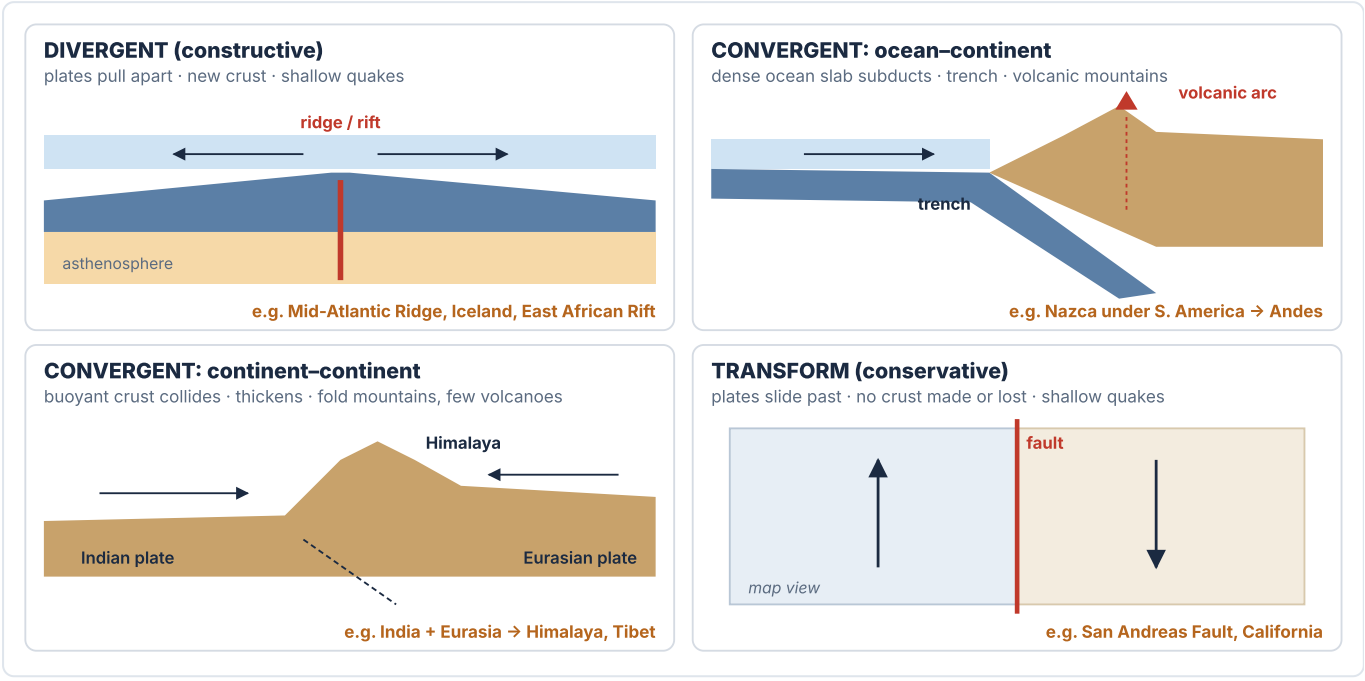


Fig. 4 — the four boundary situations. Learn one example for each; they answer every "why here?" question in this unit.

Boundary	Motion	Landforms	Quakes and volcanoes	Example
Divergent (constructive)	Apart	Mid-ocean ridge; rift valley on land	Shallow quakes; fissure (basaltic) volcanoes	Mid-Atlantic Ridge, Iceland, East African Rift, Red Sea
Convergent ocean–continent	Together; oceanic plate subducts	Trench; coastal fold mountains	Shallow to deep quakes; explosive volcanoes	Nazca under South America → Andes, Peru–Chile Trench

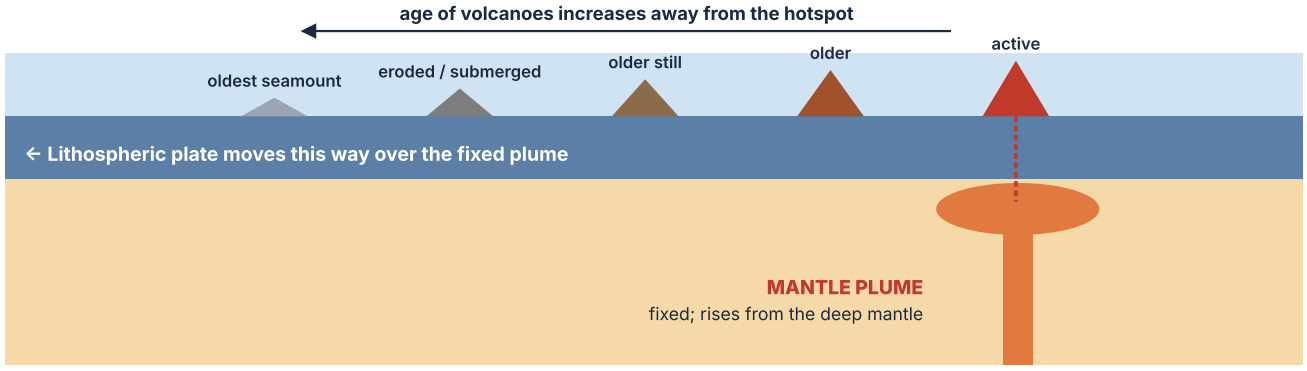
Boundary	Motion	Landforms	Quakes and volcanoes	Example
Convergent ocean–ocean	Together; older, denser plate subducts	Deep trench; volcanic island arc	Deep quakes; island-arc volcanoes	Pacific under Philippine Sea plate → Mariana Trench; Japan
Convergent continent–continent	Together; neither sinks far	Thick crust; high fold mountains and plateaux	Strong, widespread quakes; **almost no volcanoes**	India + Eurasia → Himalaya, Tibet
Transform (conservative)	Side by side	Fault lines; offset ridges	Shallow quakes; no volcanoes	San Andreas Fault

5.8 Hotspots and mantle plumes — the 2018 question

Plain version. In a few places, a narrow column of unusually hot rock rises from deep in the mantle. It stays roughly fixed while the plate above moves, so it punches a line of volcanoes through the plate, like a needle through moving cloth.

- **Mantle plume** — a narrow upwelling of abnormally hot rock from deep in the mantle, possibly from the core–mantle boundary, with a broad "head" and a thin "tail". The idea is W. Jason Morgan's (1971), building on J. Tuzo Wilson's hotspot proposal (1963).
- **Hotspot** — the area of volcanism at the surface above a plume. It is usually **inside** a plate, not at a boundary — which is why it needed its own explanation.
- **Hotspot tracks.** Because the plume is roughly fixed, volcanoes get older with distance from it. The **Hawaii–Emperor** chain is the classic case; its bend records a change in the Pacific plate's direction.
- **India's plumes.** The **Réunion** plume is linked to the **Deccan Traps**, erupted about 66 million years ago as India passed over it; its track runs through the Lakshadweep–Maldives–Chagos ridge to Réunion. The **Kerguelen** plume is linked to the **Rajmahal Traps** and the **Ninetyeast Ridge**.

Role in plate tectonics. (1) A **fixed reference frame** — hotspot tracks record the direction and speed of plate motion. (2) Plume heads can **weaken and split continents**, starting rifts (the Afar region of East Africa). (3) They build **large igneous provinces** such as the Deccan Traps, with climatic effects. (4) Some geologists see **plume push** as a minor driving force. (5) **Caveat:** whether all hotspots have deep plumes is still debated. Saying so lifts the answer.



Pacific: Hawaii–Emperor chain — its bend records a change in the Pacific plate's direction.
India: Réunion plume → Deccan Traps; Kerguelen plume → Rajmahal Traps and the Ninetyeast Ridge.

Fig. 5 — a hotspot track. Draw this for 2018-type questions; add the Réunion–Deccan example for India.

5.9 Why mountains, earthquakes and volcanoes line up — the 2014 and 2020 questions

- **Fold mountains form at convergent boundaries**, where thick sediments on continental margins are squeezed and folded. Today's convergent boundaries mostly lie along continental edges (Andes, Rockies) or along collision zones (Himalaya, Alps). So the young fold mountains are found at margins.
- **Older fold mountains are now inland** — the Appalachians, the Urals, the Aravallis. They mark the sites of ancient collisions, now welded inside continents and worn down.

- **The same boundaries host earthquakes and volcanoes.** Subduction (ocean–continent, ocean–ocean) gives both earthquakes and volcanoes — the Andes, Japan, Indonesia. Collision (continent–continent) gives earthquakes but few volcanoes, because thick continental crust does not sink and melt — the Himalaya.
- **The three global belts.** The **Circum-Pacific belt** accounts for about 81% of the world's largest earthquakes; the **Alpide belt** (Java–Sumatra–Himalaya–Mediterranean–Atlantic) for about 17% (USGS); the third is along the **Mid-Atlantic Ridge**.
- **Exceptions prove the rule.** Intraplate earthquakes (Latur, 1993) and hotspot volcanoes (Hawaii) occur away from boundaries, and need separate explanations.

The Circum-Pacific belt (2020) — characteristics to list. A horseshoe of about 40,000 km round the Pacific. Mostly **subduction margins**, with the world's deepest **trenches** (the Mariana Trench's Challenger Deep, about 11 km). Volcanic **island arcs** (Aleutians, Japan, Philippines) and **young fold mountains** (Andes, Rockies). It holds a large majority of the world's active volcanoes, often given as about 75%, and about 81% of the largest earthquakes, including **deep-focus** ones along sloping Wadati–Benioff zones. It generates most of the world's **tsunamis**. Resource side: geothermal energy and copper belts (Chile, Peru).

5.10 Islands of Indonesia and the Philippines — the 2014 question

About 17,000 islands in Indonesia and about 7,600 in the Philippines come from several processes acting together:

- **Subduction island arcs.** The Indo-Australian plate subducts under the Sunda (Eurasian) plate along the Java (Sunda) Trench. The melting slab feeds the volcanic arc of Sumatra, Java, Bali and the Lesser Sundas. In the Philippines, plates subduct from both sides — the Philippine Sea plate at the Philippine Trench in the east, and the Sunda plate at the Manila Trench in the west.
- **A drowned continental shelf.** Sumatra, Java and Borneo stand on the shallow **Sunda Shelf**. During the last ice age, lower sea levels joined them as land; rising seas afterwards cut them into islands.
- **Coral islands and reefs** grow on shallow warm platforms throughout the region.
- **Collision and accretion.** Small continental fragments and arcs have been jammed together (Sulawesi, the Banda arc).

India link: the same Sunda subduction zone continues north as the Andaman–Nicobar arc. That is why **Barren Island** (India's only active volcano), the 2004 Sumatra–Andaman earthquake and tsunami, and Great Nicobar's nearness to Sumatra all belong here.

5.11 How continents and oceans change shape — the 2025 question

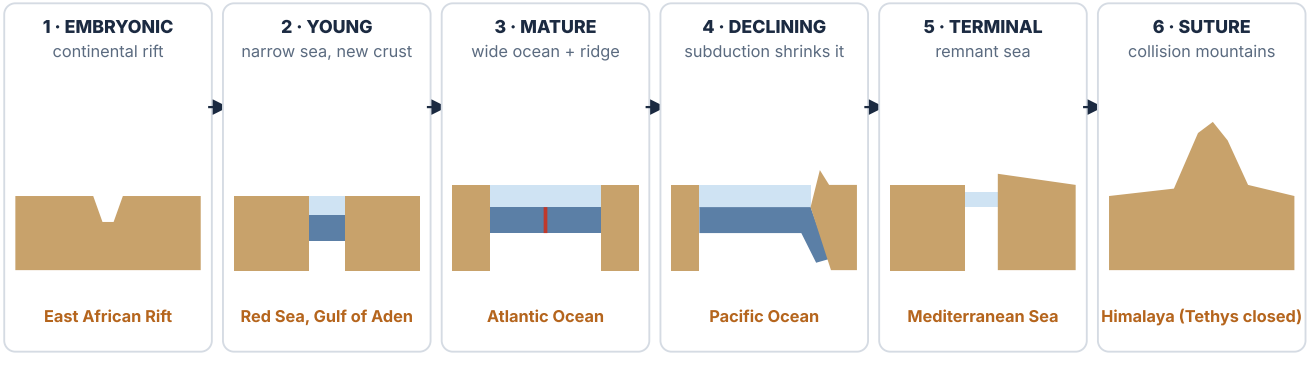


Fig. 6 — the Wilson cycle: oceans open, widen, shrink and close. Each stage has a present-day example, which is what makes this diagram score.

- **The Wilson cycle** (J. Tuzo Wilson, 1966). A continent rifts (East Africa) → a narrow sea forms (Red Sea) → it widens into an ocean (Atlantic) → subduction begins and it shrinks (Pacific) → a remnant sea remains (Mediterranean) → the continents collide and fold mountains mark the join (Himalaya, where the Tethys Sea closed).

- **The supercontinent cycle.** The same process on a larger scale: Rodinia assembled about a billion years ago and broke up; Pangaea had assembled by about 300 million years ago and began breaking up about 200 million years ago.
- **Other changes in shape and size. Accretion** — continents grow at their edges as arcs and fragments are added (western North America). **Sea-level change** — ice ages expose and drown continental shelves (the Sunda Shelf). **Isostasy** — crust rises or sinks as ice sheets or mountains load or unload it.

5.12 India as a moving plate

- India was part of **Gondwanaland** — hence the Gondwana coal and the matching fossils with Africa, Antarctica and Australia.
- After breaking away, it drifted north across the Tethys for over 100 million years — unusually fast before the collision. Palaeomagnetic evidence records the northward movement (Prelims 1995).
- It passed over the **Réunion** plume around 66 million years ago, forming the **Deccan Traps**.
- It began colliding with Eurasia about 40–50 million years ago, folding the Tethys sediments into the **Himalaya**. The Himalaya is still rising, which is why it is so earthquake-prone.
- On its eastern edge the Indian plate subducts under the Burma plate along the **Andaman–Sunda Trench** — the source of the Andaman earthquakes and of Barren Island.

6 What to memorise versus what to understand

Tier	Content
MEMORISE word for word	Layer depths: crust ~5 / ~30 / ~70 km; Moho; asthenosphere to ~400 km; Gutenberg ~2,900 km; Lehmann ~5,150 km; radius 6,371 km • P- and S-wave behaviour and the 105° / 145° shadow zones • the five drift evidences • the parent → metamorphic pairs • the boundary table with one example each • 81% / 17% (USGS) • the Wilson cycle's six stages with examples • Réunion → Deccan; Kerguelen → Rajmahal
UNDERSTAND and reproduce flexibly	Why the S-wave shadow proves a liquid outer core • lithosphere vs asthenosphere • why Wegener was rejected and what sea-floor spreading added • why collision gives quakes but no volcanoes • how a hotspot track records plate motion • why young fold mountains are at margins and old ones inland • how islands form in Indonesia and the Philippines
CONCEPTUAL recall only	The Conrad and Repetti discontinuities • the minor plates beyond Nazca and Cocos • crustal element percentages beyond the top four • the intrusive landforms beyond sill, dyke and batholith • the plume debate in detail

7 High-ROI factual anchors

Item	What to remember	Why it matters in an answer	Memory hook
Wegener, 1912–15	Continental drift; Pangaea, Panthalassa, Laurasia, Gondwana	Opening of any tectonics answer	Right idea, wrong engine
Holmes, 1930s	Mantle convection currents	The mechanism drift lacked	Boiling pot
Hess, 1961–62	Sea-floor spreading at ridges	Turns drift into a process	Conveyor belt
Vine & Matthews, 1963	Symmetric magnetic stripes	The proof of spreading	Barcode on the sea floor
McKenzie & Parker, Morgan, 1967–68	Plate tectonics	The unifying theory	Rigid on weak
Morgan, 1971; Wilson, 1963	Mantle plumes; hotspots	The 2018 answer	Needle through cloth
J. Tuzo Wilson, 1966	The Wilson cycle	The 2025 answer	Oceans breathe

Item	What to remember	Why it matters in an answer	Memory hook
USGS belts	Circum-Pacific ~81% of largest quakes; Alpide ~17%	2014 and 2020 answers	Eight in ten round the Pacific
Shadow zones	S absent beyond 105°; P gap 105°–145°	Proof of a liquid outer core	S stops, P bends
Ocean floor age	Under 200 million years; continents up to ~3,200 million	Shows oceans are recycled	Young sea, old land
Deccan Traps	Basalt; ~66 million years; Réunion plume	India example for plumes and igneous rock	India's hot-spot scar
Barren Island	India's only active volcano; Andaman Sea; erupted after 1991	Prelims 2018	Still smoking

8 Technical vocabulary bank

Term	Plain meaning	Where to use it	Example sentence
Lithosphere	Rigid outer shell: crust + top of the mantle	Any plate answer	“The lithosphere is broken into plates that ride on the asthenosphere.”
Asthenosphere	Weak, slowly flowing upper mantle	Why plates can move	“Plates glide over the ductile asthenosphere.”
Discontinuity	Boundary where seismic speed changes sharply	Interior questions	“The Moho marks the base of the crust.”
Shadow zone	Area where direct seismic waves are not recorded	Evidence of a liquid core	“The S-wave shadow beyond 105° reveals a liquid outer core.”
Subduction	One plate sinking beneath another	Trenches, arcs, volcanoes	“Subduction of the Nazca plate raised the Andes.”
Palaeomagnetism	Ancient magnetic field recorded in rocks	Sea-floor spreading; India's drift	“Palaeomagnetic data place India far south in the Mesozoic.”
Island arc	Curved chain of volcanic islands above a subduction zone	Indonesia, Japan	“Ocean–ocean convergence builds island arcs.”
Mantle plume	Narrow rising column of hot mantle rock	The 2018 question	“A plume beneath a moving plate leaves an age-progressive track.”
Large igneous province	Huge outpouring of basalt in a short time	Deccan Traps	“The Deccan Traps are a large igneous province.”
Wilson cycle	Opening and closing of an ocean basin	The 2025 question	“The Himalaya marks the last stage of the Tethys' Wilson cycle.”
Accretion	Growth of a continent by addition at its edge	Changing continent size	“Terrane accretion widened western North America.”
Isostasy	Balance of floating crust on the mantle	Uplift after ice melts	“Isostatic rebound still lifts Scandinavia.”

9 Answer expression toolkit

Definitions to reproduce

- **Plate tectonics:** the theory that the lithosphere is divided into rigid plates that move over the asthenosphere, creating crust at divergent boundaries, destroying it at convergent ones, and conserving it at transform faults.

- **Mantle plume:** a narrow, roughly stationary upwelling of abnormally hot rock from the deep mantle whose surface expression is a hotspot.
- **Primary (igneous) rock:** rock formed directly by the cooling and solidification of magma or lava; all other rocks are ultimately derived from it.
- **Sea-floor spreading:** the creation of new oceanic crust at mid-ocean ridges and its movement away from them.

Introductions

- (Tectonics) “The Earth's outer shell is a mosaic of moving plates, and most of the planet's large landforms and hazards are concentrated where these plates meet.”
- (Interior) “Since direct observation reaches barely 12 km, our picture of the interior rests mainly on the behaviour of seismic waves.”
- (Islands) “The archipelagos of South-East Asia are the product of plate convergence, sea-level change and reef growth acting together.”

Analytical and transition lines

- “Wegener supplied the evidence; Hess supplied the mechanism; plate tectonics supplied the framework.”
- “Collision builds the highest mountains but not volcanoes, because buoyant continental crust resists subduction and melting.”
- “Hotspot tracks are a geological tape recorder of plate motion.”
- “Oceans are young and recycled; continents are old and preserved.”

Conclusions

- “Plate boundaries are at once the Earth's most productive zones — of mountains, minerals and geothermal energy — and its most hazardous.”
- “The Himalaya and the Deccan Traps show that India's landscape is a record of its plate's journey.”

India linkages

Gondwana coal and fossils • Deccan Traps (Réunion) and Rajmahal Traps (Kerguelen) • Himalaya as an active collision zone • Andaman–Nicobar subduction, Barren Island and the 2004 tsunami • Makrana marble, Vindhyan sandstone, Dharwar schists as rock examples.

10 Diagram bank

Figure	Verdict	Use it for	How to draw it in about 30 seconds
Fig. 1 — interior	**Highest ROI**	Interior, waves, any “structure of the Earth” question	A quarter-circle; four bands; label Moho, Gutenberg, Lehmann with depths on the right
Fig. 2 — shadow zones	Draw for interior questions	Evidence of a liquid core	Circle with core; epicentre on top; mark 105° and 145° on both sides; shade the S gap
Fig. 3 — rock cycle	Draw for 2022-type	Rock types	Triangle of three boxes with magma above; label the arrows
Fig. 4 — boundaries	**Highest ROI**	2014, 2020, 2025 and every “why here?” question	Draw only the one you need: a trench + volcano sketch, or a collision bulge
Fig. 5 — hotspot track	Draw for 2018-type	Mantle plumes	A plate bar, a plume stem, three volcanoes getting smaller away from it
Fig. 6 — Wilson cycle	Draw for 2025-type	Changing continents and oceans	Six small boxes in a row; one example under each

Examiner's-eye view. For a 10-mark answer, one well-labelled sketch is worth more than a paragraph. Pick the single diagram that answers the demand — Fig. 4 for "why here", Fig. 5 for plumes, Fig. 6 for change over time — and spend no more than two minutes on it.

11 Comparisons

11.1 Continental vs oceanic crust

Basis	Continental	Oceanic
Thickness	~30 km average, up to ~70 km	~5 km average
Rock	Granitic (sial), lighter	Basaltic (sima), denser
Age	Up to ~3,200+ million years	Under 200 million years
Fate at convergence	Resists subduction	Subducts

11.2 Drift vs sea-floor spreading vs plate tectonics

Basis	Continental drift	Sea-floor spreading	Plate tectonics
Proposed by	Wegener, 1912–15	Hess, 1961–62	McKenzie, Parker, Morgan, 1967–68
What moves	Continents through ocean crust	Ocean floor away from ridges	Whole lithospheric plates
Force	Pole-fleeing and tidal (too weak)	Mantle convection	Convection, ridge push, slab pull
Weakness	No mechanism	Explains oceans, not continents' edges	Plume role still debated

12 Master questions and answers

MASTER ANSWER 1 (10 marks, 150 words). "Why are the world's fold mountain systems located along the margins of continents? Bring out the association between the global distribution of fold mountains and earthquakes and volcanoes." (2014)

> Fold mountains form where plates converge and compress the thick sediments that accumulate on continental margins.
>
> **Why at margins.** Most convergent boundaries today lie along continental edges. Where oceanic crust subducts beneath a continent, margin sediments are crumpled into ranges such as the Andes and Rockies. Where two continents collide, the sediments of the closing sea are folded into ranges such as the Himalaya and Alps. Older fold mountains — the Appalachians, Urals and Aravallis — mark ancient collisions now inside continents.
>
> **Association with quakes and volcanoes.** The same convergent zones are the Earth's main seismic belts: the Circum-Pacific accounts for about 81% of the largest earthquakes and the Alpide belt for about 17% (USGS). Subduction margins also melt rock and produce volcanoes (Andes, Japan), but collision zones like the Himalaya have strong earthquakes and few volcanoes, because buoyant continental crust does not sink and melt.
>
> Fold mountains, earthquakes and volcanoes are thus three expressions of one process — plate convergence.

About 165 words. Add a Fig. 4 sketch of ocean–continent convergence if time allows; cut the older-mountains sentence first if pressed.

MASTER ANSWER 2 (10 marks, 150 words). “Define mantle plume and explain its role in plate tectonics.” (2018)

> A mantle plume is a narrow column of abnormally hot rock rising from deep in the mantle, possibly from the core–mantle boundary. Its surface expression is a hotspot, usually located inside a plate rather than at a boundary.
>
> **Role in plate tectonics.** (1) **Recording plate motion** — because plumes are roughly fixed, a moving plate carries a line of volcanoes that get older away from the hotspot, as in the Hawaii–Emperor chain. (2) **Starting rifts** — a plume head can weaken and split a continent, as beneath Afar in East Africa. (3) **Large igneous provinces** — India's Deccan Traps formed as the Indian plate passed over the Réunion plume, and the Rajmahal Traps are linked to the Kerguelen plume. (4) **Plume push** may add a minor driving force.
>
> Plumes thus complement boundary processes by explaining intraplate volcanism, though whether all hotspots have deep roots is still debated.

About 150 words. Draw Fig. 5 in the margin. The final caveat is what separates it from a textbook answer.

MASTER ANSWER 3 (15 marks, 250 words) — skeleton. “Discuss how the changes in shape and sizes of continents and ocean basins of the planet take place due to tectonic movements of the crustal masses.” (2025)

- **Open:** the lithosphere is a set of moving plates, so continents and oceans are temporary arrangements, not fixed features.
- **Body 1 — the Wilson cycle** with Fig. 6: rift (East Africa) → young sea (Red Sea) → mature ocean (Atlantic) → shrinking ocean (Pacific) → remnant sea (Mediterranean) → suture and mountains (Himalaya).
- **Body 2 — the boundary processes behind it:** divergence adds ocean floor; subduction consumes it; collision joins continents; transform faults shift them.
- **Body 3 — continents grow and shrink:** accretion of arcs and fragments; rifting splits them; the supercontinent cycle (Rodinia, Pangaea).
- **Body 4 — non-tectonic modifiers:** sea-level change drowns or exposes shelves (Sunda Shelf); isostasy lifts or lowers crust.
- **India:** Gondwana break-up, drift across the Tethys, collision raising the Himalaya.
- **Close:** the Atlantic is widening and the Pacific narrowing today, so the map is still changing.

13 Mains analytical question bank

#	Question	Marks	Tags
1	Define mantle plume and explain its role in plate tectonics.	10	H R D
2	Why are fold mountains located along continental margins? Bring out their association with earthquakes and volcanoes.	10	H R D
3	Discuss the geophysical characteristics of the Circum-Pacific zone.	10	H R
4	Describe the characteristics and types of primary rocks.	10	H D I
5	Explain the formation of the islands of the Indonesian and Philippine archipelagos.	10	H D
6	Discuss how continents and ocean basins change shape and size due to tectonic movements.	15	H R D
7	How does the study of seismic waves reveal the structure of the Earth's interior?	10	R D
8	“Sea-floor spreading supplied the mechanism that continental drift lacked.” Discuss.	15	R
9	Why does the Himalaya have frequent earthquakes but no active volcanoes, while the Andes has both?	10	R D I
10	Examine the significance of the Réunion and Kerguelen hotspots in India's geological history.	10	I
11	“Plate boundaries are zones of both hazard and resource.” Examine with examples.	15	R I

H = asked before • R = high ROI • D = needs a diagram • I = needs an India linkage. Questions 7–11 are mine, built from where the PYQs point next.

14 Prelims trap list

- S-waves travel through **solids only**; P-waves through all states. “S-waves pass through the outer core” is false.
- The core is mainly **iron** (with nickel), not silicon or aluminium.
- **Clay is porous but impermeable**; “clay is not porous” is false.
- **Quartzite** comes from sandstone; **marble** from limestone; **slate** from shale.
- **Vosges** and the Black Forest are block mountains; Alps, Andes, Himalaya, Appalachians are fold mountains.
- Barren Island is **active** and has erupted several times since 1991; it lies in the Andaman Sea, not east of Great Nicobar.
- Ocean floor is **younger** than continental rock everywhere.
- Collision zones have earthquakes but **very few volcanoes**.
- Hotspots are usually **inside** plates, not at their edges.

15 Closed-book recall drill

Core concepts (5)

- Name the three layers and the boundary between each, with depths.
- Explain in two sentences how the S-wave shadow zone proves the outer core is liquid.
- Distinguish lithosphere from asthenosphere, and crust from mantle.
- Give five pieces of evidence for continental drift.
- State the four boundary situations with one example each.

Factual anchors (5)

- Who proposed sea-floor spreading, and what did Vine and Matthews add?
- Give three parent → metamorphic rock pairs.
- Which plumes made the Deccan and Rajmahal Traps?
- What share of the largest earthquakes occur in the Circum-Pacific and Alpide belts?
- Name the six stages of the Wilson cycle with examples.

Comparisons (2)

- Continental vs oceanic crust on four bases.
- Drift vs spreading vs plate tectonics on four bases.

Diagrams (2)

- Draw Fig. 1 with all depths from memory.
- Draw Fig. 5 and write the India example under it.

Timed answer (1)

- In 8 minutes, write the 2014 fold-mountain answer with one sketch.

16 One-page quick-revision sheet

THE SPINE. We infer the interior from seismic waves → the Earth is layered: crust (5/30/70 km), mantle (to 2,900 km), liquid outer core, solid inner core → a rigid lithosphere rides on a weak asthenosphere → it is broken into plates → plates diverge, converge or slide → oceans open and close in Wilson cycles.

ESSENTIAL NUMBERS
Radius 6,371 km • Moho ~5 km under oceans, ~30–70 km under continents • asthenosphere to ~400 km • Gutenberg ~2,900 km • Lehmann ~5,150 km • S shadow beyond 105°; P gap 105°–145° • ocean floor <200 Ma • Circum-Pacific ~81%, Alpide ~17% of largest quakes • Deccan Traps ~66 Ma

ESSENTIAL NAMES
Wegener 1912–15 • Holmes 1930s • Hess 1961–62 • Vine & Matthews 1963 • McKenzie & Parker 1967, Morgan 1968 • Wilson 1963 (hotspots), 1966 (cycle) • Morgan 1971 (plumes)

BOUNDARIES — Divergent: Mid-Atlantic Ridge, East African Rift • O–C: Nazca → Andes • O–O: Mariana Trench, Japan • C–C: Himalaya (quakes, no volcanoes) • Transform: San Andreas

ROCKS — Igneous: intrusive granite / extrusive basalt (Deccan) • Sedimentary: sandstone, shale, limestone, coal (Gondwana) • Metamorphic: limestone→marble (Makrana), sandstone→quartzite, shale→slate, granite→gneiss

THE THREE LINES TO HAVE WORD-PERFECT
1. S-waves cannot pass through liquid, so their absence beyond 105° proves the outer core is liquid.
2. Wegener supplied the evidence; Hess supplied the mechanism; plate tectonics supplied the framework.
3. Collision builds the highest mountains but not volcanoes, because continental crust does not subduct and melt.

CONFUSIONS TO AVOID — lithosphere ≠ crust • clay is porous but impermeable • Vosges is a block mountain • Barren Island is still active • hotspots are inside plates • Repetti and Conrad are coaching names — fine for Prelims, do not lean on them in Mains

17 Sources and honest caveats

- **PYQs** — your UnlockIAS files: Mains GS-I 2013–2026 (verbatim) and Prelims Geography 1995–2026. The 2026 Prelims tagging may be incomplete.
- **Standard sources** — NCERT Class XI *Fundamentals of Physical Geography* (interior, distribution of oceans and continents, minerals and rocks); USGS for the earthquake-belt shares.
- **Depths are rounded** conventional figures. Real boundaries vary by place; write “about”.
- **Repetti and Conrad** appear in Indian coaching material but are rarely used in current geology. Use them for Prelims recognition only.
- **The share of active volcanoes** in the Ring of Fire is often given as about 75%; it depends on how “active” is counted. Prefer “a large majority”.
- **Plume theory is debated.** The Réunion–Deccan and Kerguelen–Rajmahal links are widely accepted; whether every hotspot has a deep plume is not.
- **The G1/G2 split is mine.** Earthquake measurement, volcano types and tsunamis are in G2.